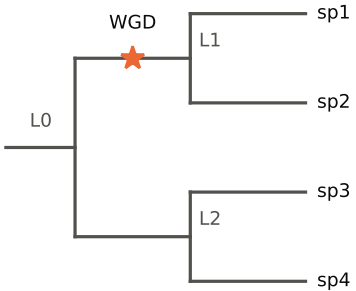
A

Literature survey	verified citations; ~85 candidate families
Paper retrieval	Europe PMC full text, supplements, human requests
Extraction by parsers	IDs, labels, trees; quotes for every event
Mapping to panels	same species; ≥98% id, ≥80% coverage, RBH
Reference tree	S / D / H / U nodes, level intervals
Verification & freeze	blind re-derivation, audit, checksums

B



Reference merge levels of gene pairs:

a1–a2 (sp1–sp2 orthologs)	merge = [L1, L1]
a1–b1 (ohnologs, WGD on L1 branch)	merge = [L0, L0]
a1–c (sp1–sp3, unresolved)	not scored
timing contested	merge = [L1, L0] interval